

BucHunt Software Requirements Specification

ScavengeRUS project / Requirements review / September 22, 2026

This document defines the participant, team, host, and administrator capabilities for the browser-based BucHunt scavenger hunt system. Requirements are organized by workflow, followed by security, usability, and data requirements. The team should use the requirement identifiers to build its backlog and acceptance tests.

MUST identifies a required capability within the stated conditions. MAY identifies an optional capability. Items awaiting decisions are listed at the end. Delivery dates, supported configurations, and test conditions must be agreed before the requirements are finalized.

1 Mandatory Functional Requirements

Account Registration and Hunt Access

FR-001: Landing Page

The system MUST provide a public landing page that allows a participant to register for an available hunt or access a hunt using the appropriate hunt URL and access code.

FR-002: Account Creation

When account creation is enabled, the system MUST allow a user to create an account using the information required by the configured deployment.

FR-003: Participant Registration

The system MUST provide a digital registration form that collects the information required to enroll a participant in a selected hunt.

FR-004: Access Code Assignment

The system MUST assign each participant or team an access code associated with the specific hunt in which they are enrolled.

FR-005: Access Code Validation

When a participant submits an access code, the system MUST verify that the code is valid and active for the requested hunt before granting access.

FR-006: Access Code Expiration

The system MUST prevent an access code from being used when the associated hunt has ended or when an authorized host has disabled the code.

Hunt Administration

FR-007: Hunt Creation

An authorized Admin MUST be able to create a hunt containing, at minimum, a title, start date/time, end date/time, URL, and task list.

FR-008: Multiple Hunts

The system MUST maintain separate configurations, participants, tasks, access codes, and progress records for multiple hunts.

FR-009: Concurrent Hunts

When concurrent hunts are enabled for the deployment, the system MUST allow multiple hunts to operate simultaneously without mixing participant, task, or progress data between hunts.

FR-010: Hunt Status

The system MUST identify each hunt using a defined status such as Pending, Active, or Expired based on its configured state and dates.

Businesses and Locations**FR-011: Add Participating Location**

An authorized business, Admin, or host MUST be able to add a participating location containing the information required by the hunt, including a display name and geographic coordinates when the location is used for GPS validation.

FR-012: Location Updates

When an authorized location is added, modified, removed, or activated, the participant-facing hunt information MUST reflect the change on the next successful refresh or synchronization.

FR-013: Hunt Map

The system MUST display the locations associated with the participant's current hunt on a map.

FR-014: Map Directions

A participant MUST be able to select a hunt location from the map and access directions or routing information to that location through the configured mapping capability.

FR-015: BucHunt GPS Interface

For BucHunt, participant-facing GPS/location functionality MUST be presented through a map-based interface.

Tasks**FR-016: Task List**

After successfully entering a hunt, a participant MUST be able to view the tasks assigned to that hunt.

FR-017: Task Ordering

The participant task view MUST display unfinished tasks before completed tasks.

FR-018: Start Task

The system MUST allow a participant to start any task that is currently available according to the hunt's task-order rules.

FR-019: Complete Task

When a participant successfully satisfies the validation rules for a task, the system MUST mark that task as completed for the participant or team.

FR-020: Completion Timestamp

For every successfully completed task, the system MUST store the task identifier, participant/team identifier, hunt identifier, and completion timestamp on the server.

FR-021: Submission Result

After a task submission is processed, the system MUST inform the participant whether the submission was valid, invalid, or unable to be validated.

FR-022: Hunt Completion Message

When a participant or team successfully completes all required tasks in a hunt, the system MUST display a congratulations message.

QR Codes**FR-023: Unique QR Codes**

Each QR code used for task validation MUST be uniquely associated with the appropriate task/hunt so that a QR code from one hunt cannot incorrectly complete a task in another hunt.

FR-024: Location QR Codes

The system MUST support QR codes displayed at participating locations to identify the location and/or support hunt task interaction.

FR-025: QR Scanning

When QR validation is enabled for a task, the participant MUST be able to use a supported device camera to scan the task's QR code and submit its encoded value for validation.

FR-026: Manual QR Entry

For a QR-based task, the system MUST provide a human-readable value that can be manually entered when QR scanning is unavailable or unsuccessful.

FR-027: QR Validation

When a QR value is submitted, the system MUST compare it with the value configured for the participant's current hunt/task and MUST reject values that do not match.

GPS and Location Validation**FR-028: GPS Task Validation**

When GPS validation is enabled for a task and the participant grants location permission, the system MUST compare the participant's current location with the task's stored latitude and longitude.

FR-029: GPS Completion Distance

For tasks using the Tier 1 location rule, the system MUST accept GPS completion only when the participant is within 50 feet of the configured task location.

FR-030: GPS Permission Failure

If GPS permission is denied or the participant's location cannot be obtained, the system MUST inform the participant that GPS validation cannot currently be performed and MUST NOT record the task as GPS-completed.

Teams and Progress**FR-031: Team Access**

The system MUST allow multiple participants to use the same team access code concurrently when the access code represents a team.

FR-032: Shared Team Progress

When one team member completes a task, the completion MUST be recorded as progress for the team.

FR-033: Cross-Device Synchronization

When team progress changes, concurrently connected team devices **MUST** receive the updated progress within the synchronization limit defined by NFR-013.

FR-034: Progress Calculation

The system **MUST** calculate participant/team progress using the number of successfully completed required tasks and, when enabled, the achievement points assigned to those tasks.

FR-035: Ranking

When ranking is enabled, a participant **MUST** be able to view their ranking relative to eligible participants or teams in the same hunt.

Communication and Privacy**FR-036: Participant Messaging**

An authorized host **MUST** be able to request an application-mediated message to selected or all enrolled participants in a hunt without receiving direct access to protected participant contact information solely for that purpose.

FR-037: Communication Preferences

The system **MUST** allow participants to configure the supported communication opt-out preferences, including sponsor, hunt-organization, and application communications where applicable.

FR-038: Opt-Out Enforcement

Before sending an optional host, sponsor, or platform communication, the system **MUST** check the participant's applicable communication preferences and **MUST** suppress messages the participant has opted out of receiving.

2 Optional Functional Requirements**FR-039: Electronic Access Code Delivery**

The system **MAY** deliver a participant's access code through email, SMS, or both when the appropriate contact information and messaging service are available.

FR-040: Multiple Hunt Enrollment

The system **MAY** allow a participant to participate in multiple hunts simultaneously while maintaining separate access credentials and progress for each hunt.

FR-041: White-Label Branding

The system **MAY** allow an authorized host to configure supported branding elements such as hunt name, logo, colors, introduction text, and theme.

FR-042: Task Presentation Modes

The system **MAY** allow a host to configure a hunt's tasks as fixed-order, randomized, or incremental.

FR-043: Multiple Task Types

The system **MAY** support location, multiple-choice trivia, single-word trivia, image collection, QR-code scanning, and combinations of supported validation methods.

FR-044: Points Display

When achievement points are enabled, the system MAY display the participant's earned points and total possible points.

FR-045: Shared Device Teams

Team members MAY participate using separate supported devices or a shared device.

FR-046: Live Ranking Updates

When live rankings are enabled, the system MAY automatically update displayed rankings during an active hunt.

FR-047: Prize Criteria

The system MAY allow an authorized host to define prize criteria using measurable hunt data such as tasks completed, locations visited, completion time, points, or configured participant categories.

FR-048: Hunt Results

When prize criteria are configured, the system MAY retain the hunt result information necessary to determine qualifying recipients after the hunt ends.

3 Mandatory Nonfunctional Requirements

Security and Privacy

NFR-001: Encryption at Rest

Sensitive participant information stored on the server MUST be encrypted at rest using the encryption mechanism approved for the deployment environment.

NFR-002: Encryption in Transit

Production communication containing authentication credentials, participant information, access codes, or other sensitive information MUST use HTTPS/TLS.

NFR-003: Authorization

The system MUST enforce authorization checks before providing access to hunt, participant, administrative, location, or protected data.

NFR-004: Hunt Data Isolation

A user MUST NOT be able to retrieve protected data belonging to a hunt for which their authenticated role is not authorized.

NFR-005: Protected Contact Information

The system MUST NOT expose a participant's protected email address or phone number to hosts or sponsors solely to allow those users to send application-mediated messages.

NFR-006: Secrets Management

Production API keys, passwords, private keys, and other application secrets MUST NOT be committed to the source-code repository or delivered to browser clients when the secret is intended only for server use.

NFR-007: Safe Logging

Application logs MUST NOT contain plaintext passwords, authentication secrets, or other credentials. Sensitive participant information SHOULD be excluded or appropriately protected when logging is operationally necessary.

Compatibility and Usability

NFR-008: Browser-Based Interface

Core participant functionality **MUST** be accessible through a web browser without requiring installation of a native mobile application.

NFR-009: Device Support

Core participant workflows **MUST** remain usable on supported modern desktop browsers, smartphones, and tablets.

NFR-010: Permission Handling

When camera or location permission is denied, unavailable, or revoked, the application **MUST** display an understandable error/state and **MUST** remain usable for functionality that does not require that permission.

Data and Synchronization

NFR-011: Server-Side Persistence

Hunt configuration, player progress, and task-completion records **MUST** be stored on the server and **MUST NOT** depend on browser cookies as the authoritative data store.

NFR-012: Hunt Isolation

Running multiple hunts simultaneously **MUST NOT** cause access codes, tasks, player progress, or protected participant information from one hunt to be incorrectly associated with another hunt.

NFR-013: Synchronization Performance

Team progress updates **MUST** become visible on concurrently connected team devices within 5 seconds under normal operating conditions.

The 5-second target is proposed and requires team approval. The test workload and network conditions must be defined before acceptance testing.

4 Optional Nonfunctional Requirements

NFR-014: Administrative Audit Log

The system **MAY** record privileged administrative actions involving hunt configuration, participant tracking, permissions, or protected information, including the acting user, action, affected resource, and timestamp.

NFR-015: Third-Party Failure Recovery

The system **MAY** recover from temporary failures of mapping, email, SMS, or other external services without corrupting hunt or participant progress.

NFR-016: Extensibility

The system **MAY** be structured so additional task types and validation methods can be added without requiring redesign of unrelated hunt and participant functionality.

NFR-017: Accessibility

The participant-facing interface **MAY** target WCAG 2.2 Level AA accessibility where practical for the project scope.

NFR-018: Production Performance Targets

Before production deployment, the team MAY establish measurable targets for response time, concurrent-user capacity, backup frequency, restoration time, and recovery objectives.